AP PRECALCULUS

Unit 1 • Lesson 1.2 notes

Lesson 1.2 • Rates of Change

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

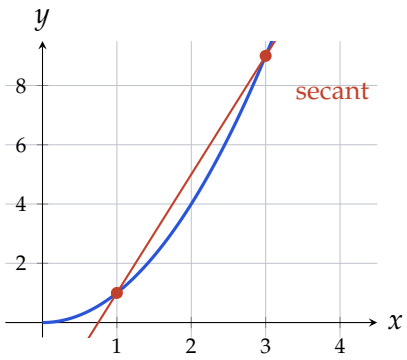
- Compute and interpret the average rate of change of a function on an interval.
- Relate the average rate of change to the slope of a secant line.
- Estimate the rate of change at a point using small intervals.

Vocabulary. average rate of change, secant line, slope, rate of change at a point, units.

1. Average rate of change

Key idea

The average rate of change of f on $[a,b]$ is $\dfrac{f(b)-f(a)}{b-a}$, the change in output divided by the change in input. It equals the slope of the **secant line** through $(a,f(a))$ and $(b,f(b))$.



For $f(x)=x^2$ on $[1,3]$, the secant line has slope $\frac{9-1}{3-1}=4$.

2. Units and interpretation

The units of an average rate of change are *output units per input unit*. A complete interpretation names the quantity, the interval, and the units.

Example 1 • Interpreting

The number of visitors $V(t)$ to a website, t hours after midnight, satisfies $V(2)=150$ and $V(6)=870$.

Solution. $\frac{870-150}{6-2}=180$. Between 2 a.m. and 6 a.m., the number of visitors increased at an average rate of 180 visitors per hour.

3. Rate of change at a point

To estimate how fast f changes at one input a, compute average rates of change on smaller and smaller intervals around a.

Example 2 • Estimating

Estimate the rate of change of $f(x) = x^3$ at $x = 2$.
Solution. $[2, 2.1]$: $\frac{9.261-8}{0.1} = 12.61$. $[2, 2.01]$: $\frac{8.120601-8}{0.01} = 12.0601$. $[1.99, 2]$: $\frac{8-7.880599}{0.01} = 11.9401$. The rate of change at $x = 2$ is about 12.

Sign of the rate of change

Positive rate of change: increasing. Negative: decreasing. If f changes from increasing to decreasing at $x = a$, its rate of change changes from positive to negative there.

Common mistakes

- Computing $\frac{b-a}{f(b)-f(a)}$ (upside down), or subtracting in different orders: $\frac{f(b)-f(a)}{a-b}$.
- Forgetting units, or giving an average rate of change as if it were the change itself.

AP exam tip

"Average rate of change" questions on the exam often ask for an interpretation: use the template *"On average, [quantity] [increases/decreases] by [number] [output units] per [input unit] from [input a] to [input b]."*

Quick check

1. Find the average rate of change of $g(x) = 5 - x^2$ on $[-1, 2]$.
2. $T(4) = 21$ and $T(10) = 9$, where T is a temperature in °C and t is in hours. Interpret $\frac{T(10)-T(4)}{10-4}$.

Answers. 1. $\frac{1-4}{3} = -1$. 2. -2: from hour 4 to hour 10 the temperature dropped on average by 2 °C per hour.